Final long and complex URL stress test

2,200 frozen URLs • 9.09% phishing • unchanged models and thresholds

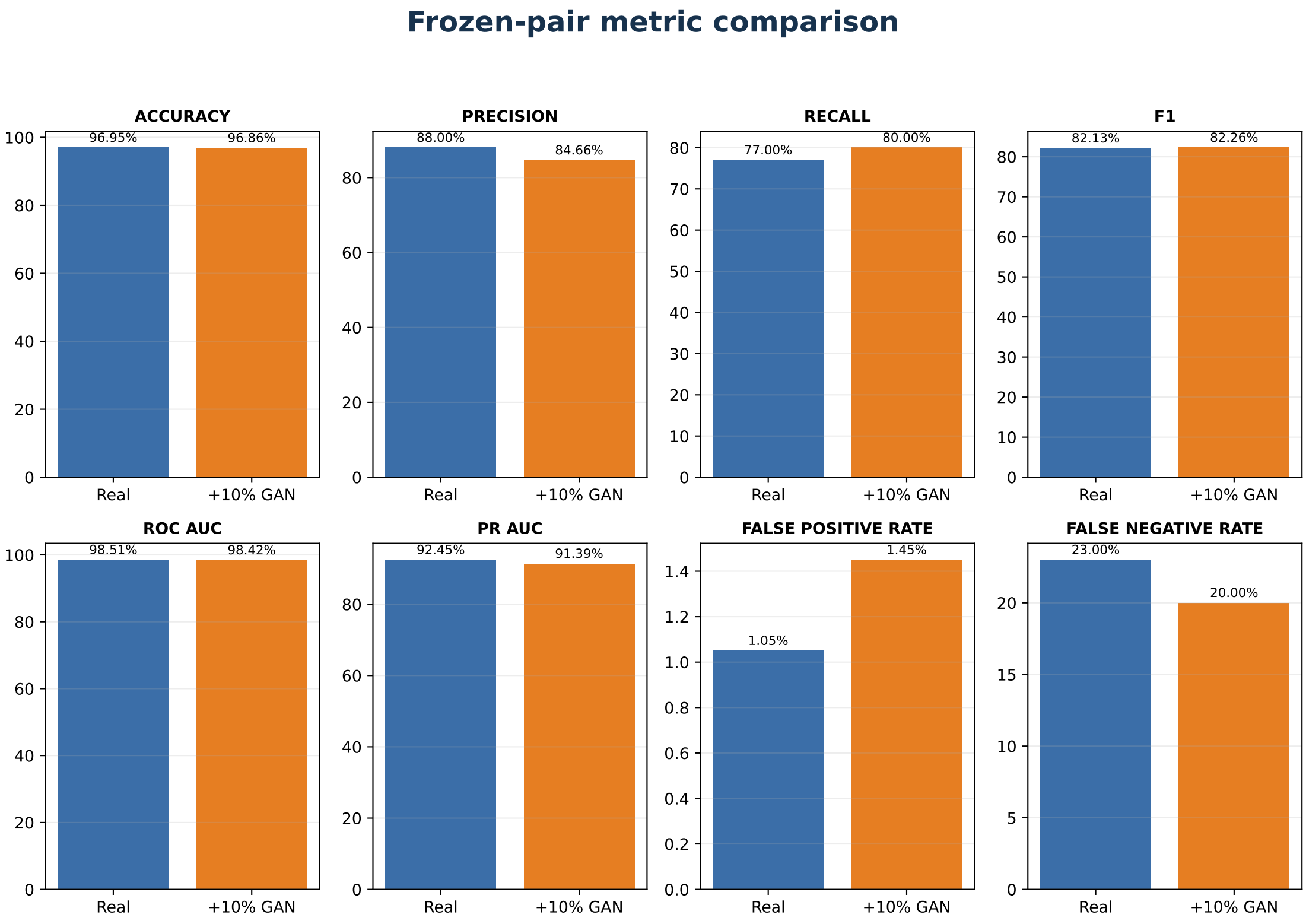
Condition	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC	FPR	FNR
HYB-MS real-only	96.95%	88.00%	77.00%	82.13%	98.51%	92.45%	1.05%	23.00%
HYB-MS +10% GAN	96.86%	84.66%	80.00%	82.26%	98.42%	91.39%	1.45%	20.00%

LONG-URL NON-REGRESSION FAILED

GAN augmentation increased false positives on long legitimate URLs.

Paired exact McNemar p = 0.8714; GAN uniquely corrected 18 rows and uniquely worsened 20 rows.

No model, checkpoint, calibration value, threshold, or URL was changed after inference began.


Frozen-pair metric comparison
ACCURACY
96.95%
96.86%
PRECISION
88.00%
84.66%
RECALL
77.00%
80.00%
F1
82.13%
82.26%
ROC AUC
98.51%
98.42%
PR AUC
92.45%
91.39%
FALSE POSITIVE RATE
1.05%
1.45%
FALSE NEGATIVE RATE
23.00%
20.00%
Real
+10% GAN